

# E-In Son

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## RESEARCH INTERESTS

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My research focuses on learning-based autonomous navigation, with particular interest in visual navigation, traversability estimation, and representation learning. Building on earlier work in unstructured, off-road environments, I am extending toward more general navigation and robotics, including embodied agents such as navigation world models and vision-language-action (VLA) models.

## EDUCATION

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**Seoul National University** **Mar 2021 – Present**  
*Ph.D. Student, Dept. of Electrical & Computer Engineering* *Seoul, Korea*  
Vehicle Intelligence Laboratory. (Advisor: Prof. Seung-Woo Seo)

**Konkuk University** **Mar 2015 – Feb 2021**  
*B.S., Electrical & Electronics Engineering* *Seoul, Korea*  
*Summa Cum Laude* (GPA 4.33 / 4.5).

## PUBLICATIONS

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- [1] **Ordinal Neural Collapse as a Representation Prior for Visual Navigation.**  
E-In Son<sup>\*</sup>, Jung-Taak Kim<sup>\*</sup>, Seung-Woo Seo.  
*European Conference on Computer Vision (ECCV)*, 2026. [\[Paper\]](#)
- [2] **PTS-Map: Probabilistic Terrain State Map for Uncertainty-Aware Traversability Mapping in Unstructured Environments.**  
Dong-Wook Kim, E-In Son, Chan Kim, Ji-Hoon Hwang, Seung-Woo Seo.  
*IEEE Robotics and Automation Letters (RA-L)*, 2025. [\[Paper\]](#)
- [3] **Follow the Footprints: Self-supervised Traversability Estimation for Off-road Vehicle Navigation based on Geometric and Visual Cues.**  
Yurim Jeon, E-In Son, Seung-Woo Seo.  
*IEEE International Conference on Robotics and Automation (ICRA)*, 2024. [\[Paper\]](#)
- [4] **Adaptive Robot Traversability Estimation Based on Self-Supervised Online Continual Learning in Unstructured Environments.**  
Hyung-Suk Yoon, Ji-Hoon Hwang, Chan Kim, E-In Son, Se-Wook Yoo, Seung-Woo Seo.  
*IEEE Robotics and Automation Letters (RA-L)*, 2024. [\[Paper\]](#)
- [5] **Traversability-aware Adaptive Optimization for Path Planning and Control in Mountainous Terrain.**  
Se-Wook Yoo<sup>\*</sup>, E-In Son<sup>\*</sup>, Seung-Woo Seo.  
*IEEE Robotics and Automation Letters (RA-L)*, 2024. [\[Paper\]](#)

## AWARDS & HONORS

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**Earth Rovers Challenge (1st Place)** Oct 2024  
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

**Academic Excellence Scholarship** 2018 – 2020  
Konkuk University

## RESEARCH PROJECTS

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**Challengeable Future Defense Technology R&D Program (Phase 2)** 2024 – 2026

Agency for Defense Development (ADD)

*Leading the planning stack, from learning-based path prediction to real-time safe replanning, for an end-to-end visual navigation system operating in unknown, hazardous terrain.*

**Challengeable Future Defense Technology R&D Program (Phase 1)** 2022 – 2023

Agency for Defense Development (ADD)

*Developed the path tracking module for accurate and stable trajectory following.*

## TEACHING EXPERIENCE

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**Seoul National University**, Teaching Assistant

SNU in Silicon Valley Program, joint with UC Berkeley (*Summer 2024*)

Fundamentals of Random Variables and Probability Processes (*Spring 2023*)

Network Protocol and Design (*Fall 2021, 2023*)

## SKILLS

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**Tools and Languages** Python, C/C++, PyTorch, ROS

**Communication** Korean (native), English

## REFERENCE

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**Advisor:** Prof. Seung-Woo Seo

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[Website](#)